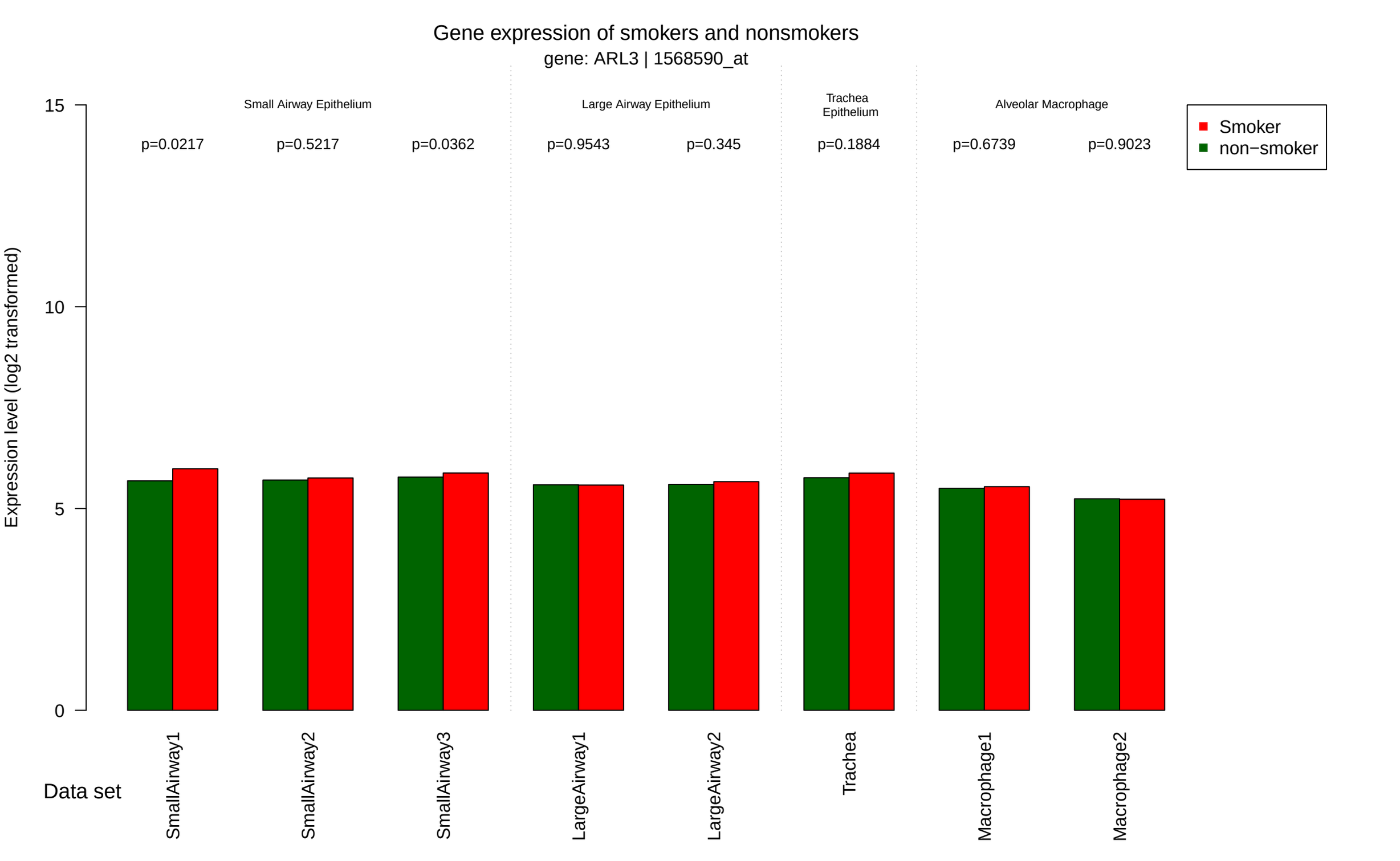
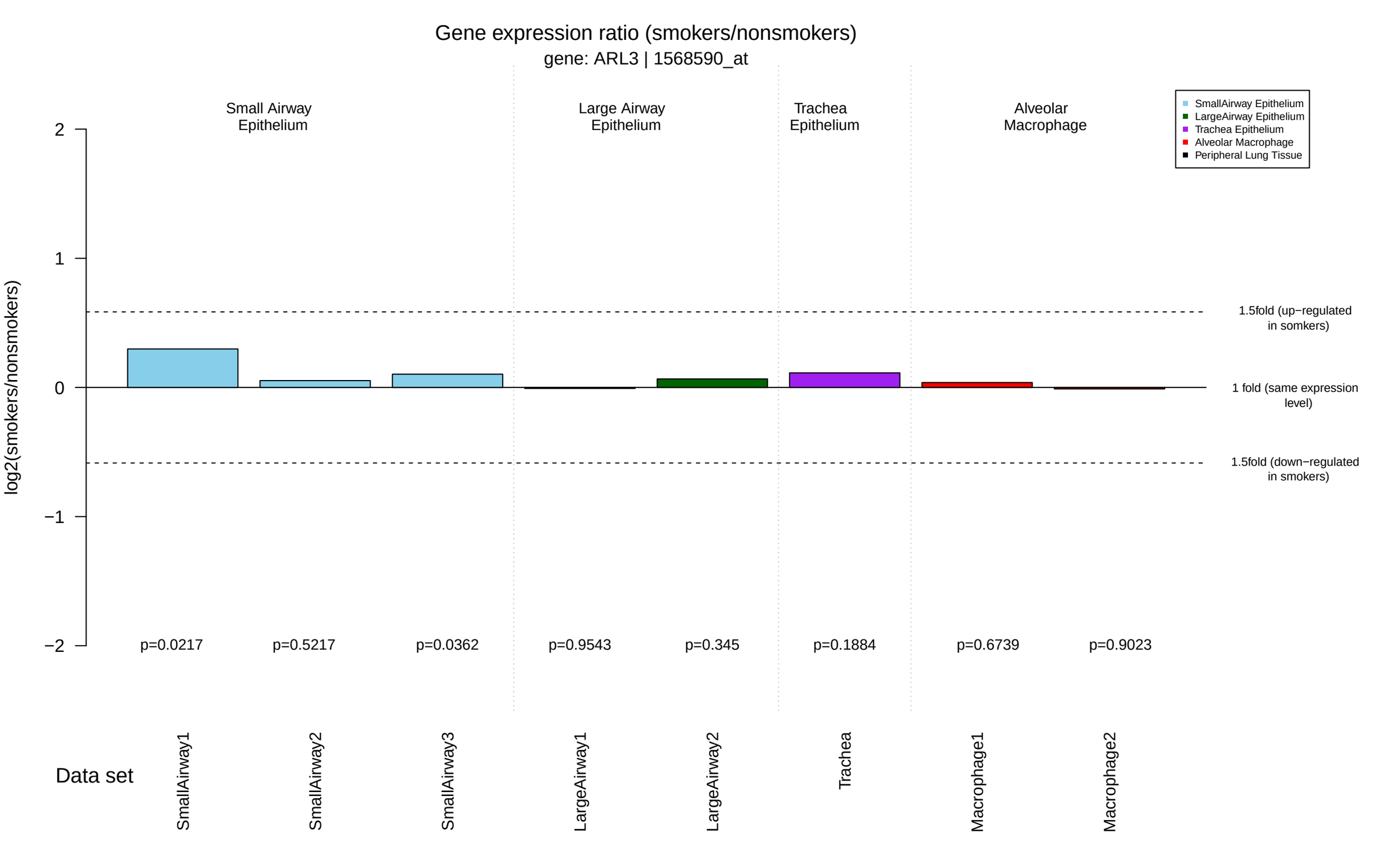
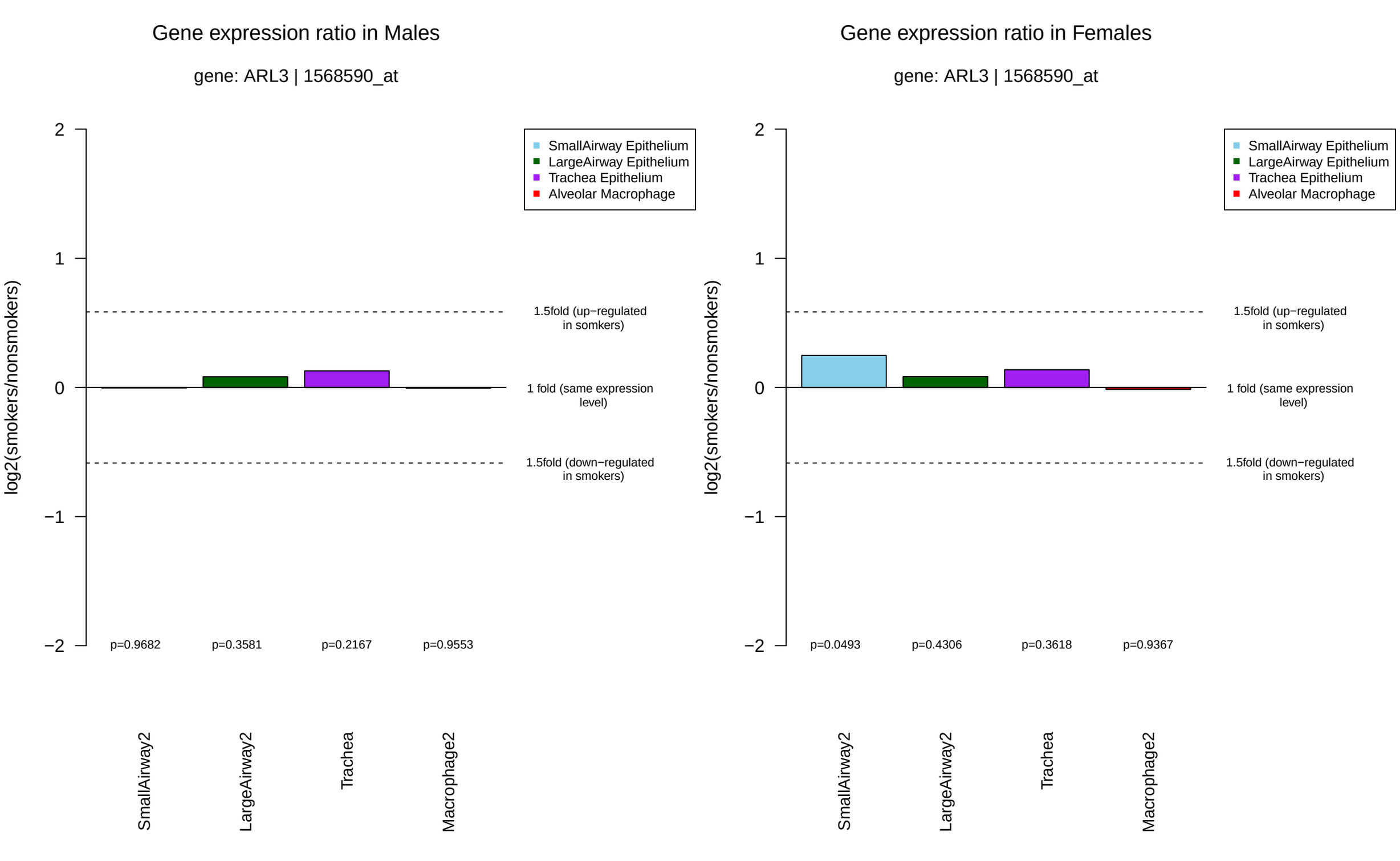
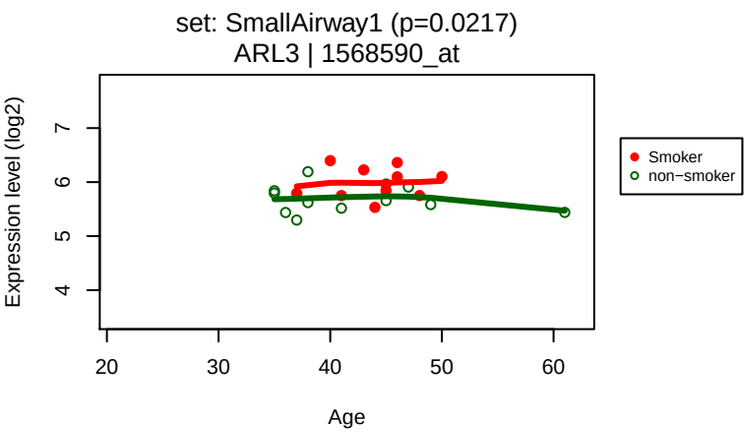
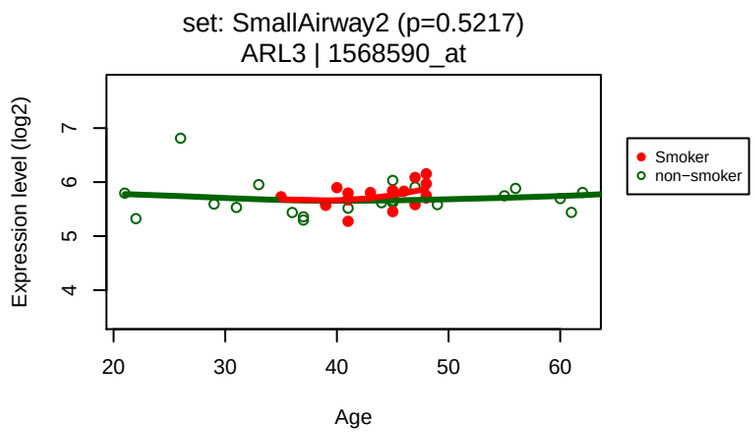
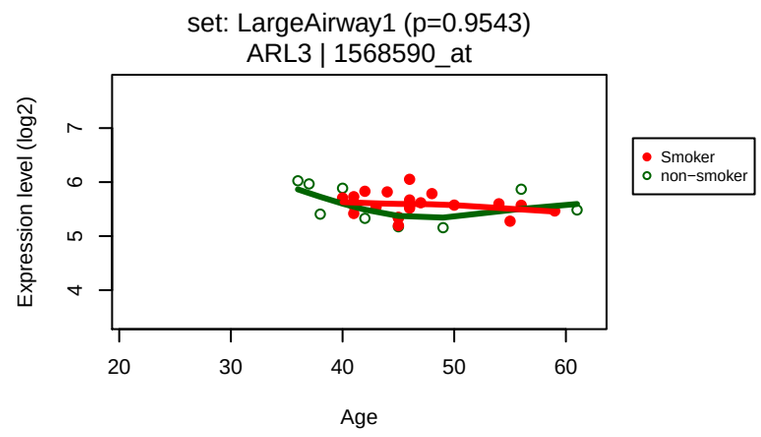

Gene expression of smokers and nonsmokers
gene: ARL3 | 1568590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0217
p=0.5217
p=0.0362
p=0.9543
p=0.345
p=0.1884
p=0.6739
p=0.9023
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

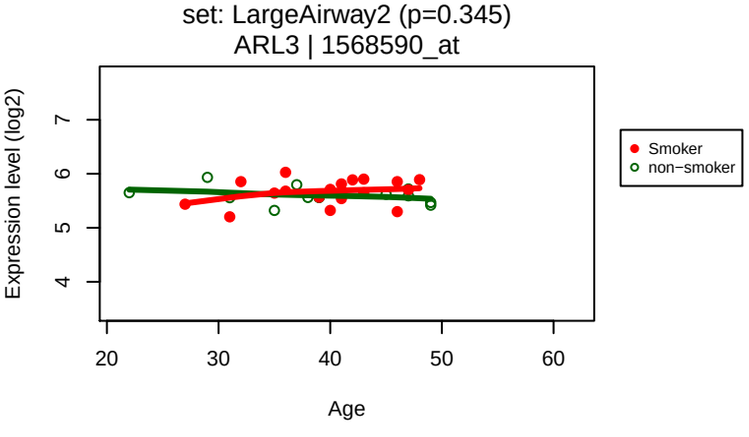

Gene expression ratio (smokers/nonsmokers)
gene: ARL3 | 1568590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0217
p=0.5217
p=0.0362
p=0.9543
p=0.345
p=0.1884
p=0.6739
p=0.9023
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

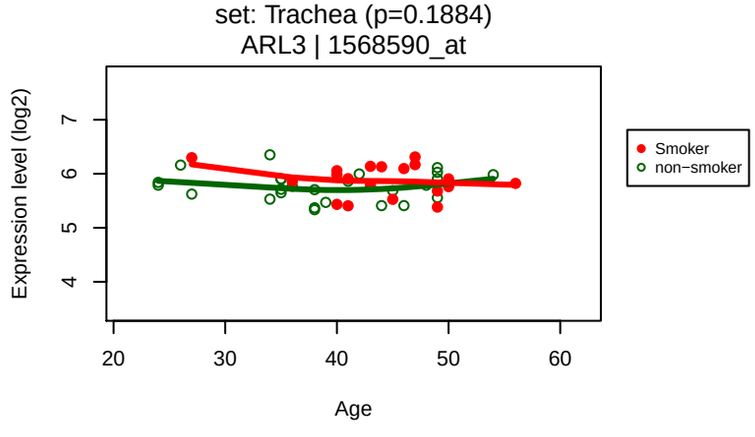

Gene expression ratio in Males
gene: ARL3 | 1568590_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9682
p=0.3581
p=0.2167
p=0.9553
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL3 | 1568590_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0493
p=0.4306
p=0.3618
p=0.9367
SmallAirway2
LargeAirway2
Trachea
Macrophage2

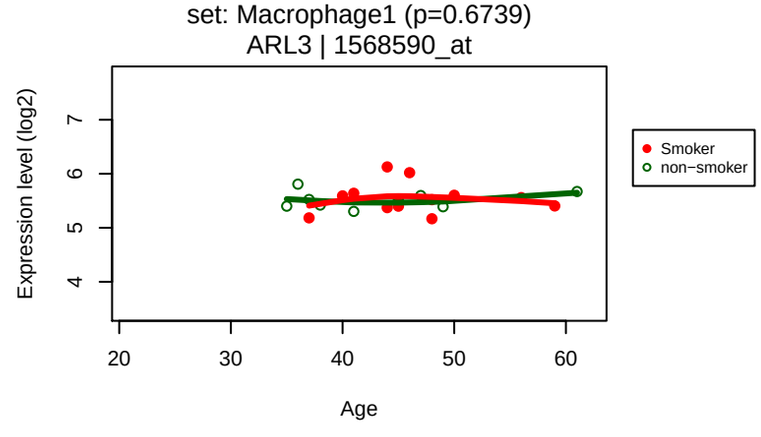

set: SmallAirway1 (p=0.0217)
ARL3 | 1568590_at
Expression level (log2)
Age
Smoker
non−smoker

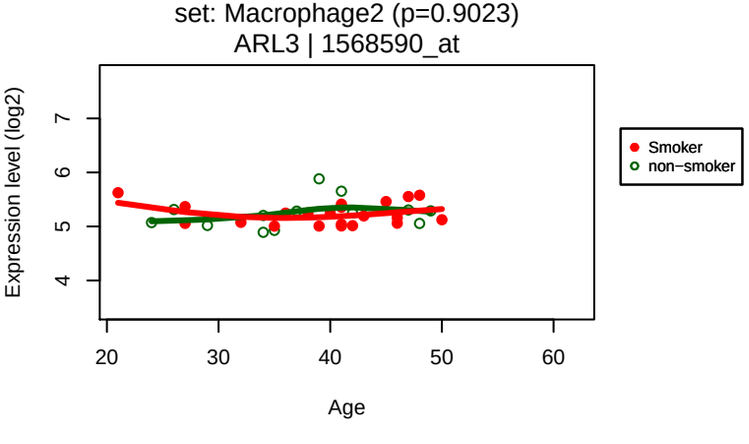

set: SmallAirway2 (p=0.5217)
ARL3 | 1568590_at
Expression level (log2)
Age
Smoker
non−smoker

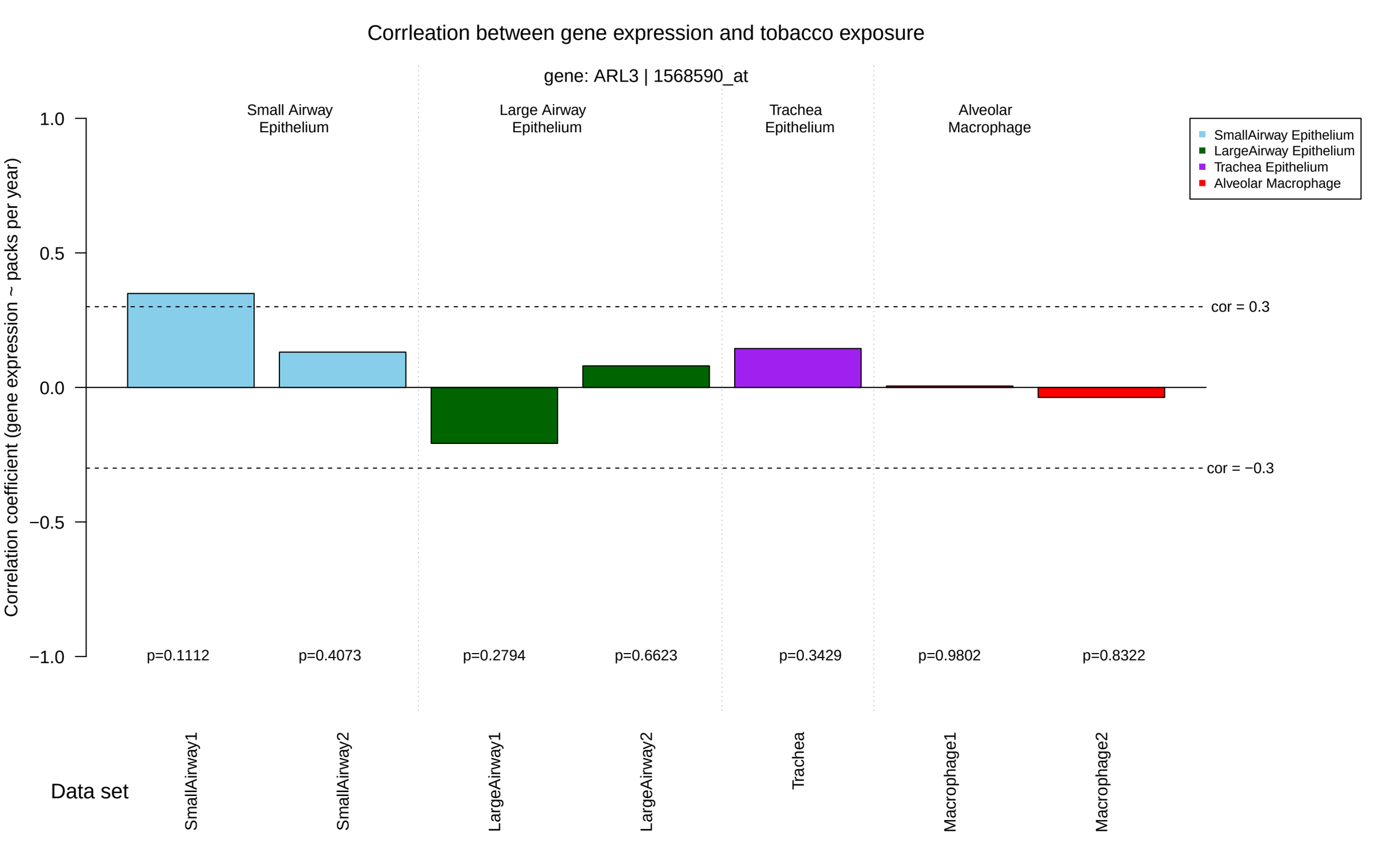

set: LargeAirway1 (p=0.9543)
ARL3 | 1568590_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.345)
ARL3 | 1568590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1884)
ARL3 | 1568590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6739)
ARL3 | 1568590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9023)
ARL3 | 1568590_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL3 | 1568590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1112
p=0.4073
p=0.2794
p=0.6623
p=0.3429
p=0.9802
p=0.8322
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2